

SQL Question and Answer –

Q1. What is the total revenue generated by male vs. female customers?

```
SELECT gender, SUM(purchase_amount) as revenue
FROM customer_data
GROUP BY gender;
```

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id, purchase_amount
FROM customer_data
WHERE discount_applied = 'Yes' AND purchase_amount >= (SELECT AVG(purchase_amount)
FROM customer_data);
```

Q3. Which are the top 5 products with the highest average review rating?

```
SELECT item_purchased, ROUND(AVG(review_rating), 2) as 'Avg Product Rating'
FROM customer_data
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Q4. Compare the average purchase amount between standard and express shipping?

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS "average purchase amount"
FROM customer_data
WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;
```

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers?

```
SELECT subscription_status, COUNT(customer_id) AS 'total_customer',
ROUND(AVG(purchase_amount), 2) AS 'Avg_purchase_amount',
ROUND(SUM(purchase_amount), 2) AS 'total_purchase_amount'
FROM customer_data
GROUP BY subscription_status
ORDER BY Avg_purchase_amount,total_purchase_amount DESC;
```

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
SELECT item_purchased, SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0
END)/COUNT(*) * 100 AS discount_rate
FROM customer_data
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Q8. What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
    SELECT category, item_purchased,
    COUNT(customer_id) AS total_order,
    ROW_NUMBER() OVER(PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS
item_rank
    FROM customer_data
    GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_order
FROM item_counts
WHERE item_rank <= 3;
```

Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer_data
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Q10. What is the revenue contribution of each age group

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer_data
GROUP BY age_group
ORDER BY total_revenue DESC;
```